Midterm Project

Start Assignment

- Due Sunday by 11:59p.m.
- Points 100
- Submitting a text entry box or a file upload
- Available until Mar 18 at 11:59p.m.

Note: This full assignment description was edited with help from Claude Opus 4.6.

Overview

In this project, you will apply the multivariate analysis toolkit from Weeks 1–8 to discover and rigorously characterize the **latent structure** underlying a real-world dataset.

Project 1 emphasized exploration and communication—telling a story about what the data reveals. Project 2 shifts the emphasis to **methodological rigour**: justifying your analytical choices, testing assumptions, quantifying uncertainty, and comparing what different methods reveal about the same data. Good visualization and clear writing remain important, but they are now expected skills rather than the primary focus.

Your central question is no longer just "What structure is there?" but **"What is the appropriate analytical framework for this data, and what does it reveal?"** Answering this requires understanding the assumptions behind each method, checking whether they hold, and interpreting results in light of those assumptions.

Datasets

You will analyze **one** of the following four datasets, assigned via ranked-choice lottery. Each dataset has properties suited to Factor Analysis and ICA, but in different ways—some are textbook FA datasets, others challenge the assumptions.

Dataset	Domain	Observations	Features
Big Five Personality	Psychology	19,718	50
Human Activity Recognition	Sensors	10,299	33
Spotify Audio Features	Music	114,000	9
World Governance Indicators	Political Science	5,045	6

Big Five Personality: Responses to the 50-item IPIP Big Five personality inventory (10 items per trait: Extraversion, Agreeableness, Conscientiousness, Neuroticism, Openness). All items on a 1–5 Likert scale. A classic psychometric dataset with a well-established theoretical factor structure.

Human Activity Recognition (HAR): Mean accelerometer and gyroscope signals from smartphones worn by 30 volunteers performing 6 activities (walking, sitting, standing, etc.). 33 features capturing time- and frequency-domain signal characteristics.

Spotify Audio Features: Audio characteristics for over 114,000 tracks across 114 genres. Features include danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo. Mixed scales (some 0–1, loudness in dB, tempo in BPM).

World Governance Indicators: World Bank panel data for 214 countries over 1996–2023. Six indicators: control of corruption, government effectiveness, political stability, regulatory quality, rule of law, and voice/accountability. All measured on a –2.5 to +2.5 scale.

Full documentation (variable descriptions, citations, loading code) is available on Quercus.

What You Need to Do

Your goal is to **discover latent structure in your dataset using methods from Weeks 1–8, justify your analytical choices, and communicate your findings with statistical rigour**. Your report will be evaluated on three things:

1. **Visualization & Communication** — Are your figures clear, well-designed, and your writing accessible?
2. **Methodological Rigour** — Do you justify method choices, test assumptions, and assess robustness?
3. **Interpretation of Structure** — Do you identify, interpret, and compare latent structure meaningfully?

The Weeks 1–8 Toolkit

You are expected to draw from the full toolkit developed over the first eight weeks. Not every method will be relevant for every dataset, but your choices—including what you *don't* use—must be justified.

Method	Weeks	Expectation
Exploratory Data Analysis	1	Expected for every submission
MVN Assessment & Hypothesis Testing	2–3	Expected: test assumptions, report and interpret results
PCA	4–5	Expected as a baseline for comparison with FA/ICA
CCA	6	Where the data has natural variable blocks
Multivariate Regression	6	Where predicting one variable set from another is meaningful
Factor Analysis	7	At least one of FA or ICA is required; justify your choice
ICA	8	At least one of FA or ICA is required; justify your choice

You must use at least one of FA or ICA, and you should explain why you chose it (or both) for your data. If you use only one, discuss why the other is or isn't appropriate.

Guiding Questions

Your analysis should address questions like:

- **Data and assumptions:** What does this data contain? What distributional properties does it have? Do the assumptions required by your chosen methods hold? (Test them.)
- **Method selection:** Why is FA, ICA, or both appropriate for this dataset? What assumptions does each make, and are they satisfied? How does PCA compare as a baseline?
- **Latent structure:** How many latent dimensions underlie the data? Justify your choice using parallel analysis, scree plots, and interpretability. What do the factors/components represent? Name them based on loading patterns.
- **Rotation and interpretation:** Does rotation improve interpretability? Compare rotated vs unrotated solutions. How do you interpret cross-loadings and communalities?
- **Relationships between variable groups:** If your data has natural blocks of variables, does CCA or multivariate regression reveal meaningful relationships between them?
- **Robustness:** How sensitive are your results to methodological choices (number of factors, rotation method, correlation type)? Do findings hold across subsamples or bootstrap replicates?

You have flexibility in how you approach these questions. Not every question will be equally relevant for every dataset. Use your judgment about what is most informative for *your* data—but justify that judgment.

What Makes a Strong Submission

Visualization & Communication: Figures are well-designed with clear labels, legends, and readable text. Captions are informative—they explain what the figure shows *and* what it means. The report reads as a coherent narrative, not a list of outputs. A reader unfamiliar with your dataset understands what it contains and what you found.

Methodological Rigour: Every major analytical choice is justified. You don't just run FA—you explain *why* FA is appropriate (or acknowledge its limitations). Assumptions are tested, not assumed. Parallel analysis is used to determine the number of factors, not just the Kaiser criterion. Results are assessed for robustness (e.g., sensitivity to the number of factors, rotation method, or subsample). Where relevant, uncertainty is quantified (e.g., bootstrap confidence intervals on loadings or communalities).

Interpretation: You identify meaningful latent dimensions and name them based on loading patterns. You connect factor structure to domain knowledge (e.g., "Factor 1 loads on all sociability items, consistent with the Extraversion dimension in personality psychology"). You compare what different methods reveal—PCA vs FA, FA vs ICA, rotated vs unrotated—and discuss what each comparison tells you. You recognize when a method isn't working well and explain why.

Methods Scope

Use Weeks 1–8 material only. Methods or visualizations not explicitly covered in Weeks 1–8 will be ignored during grading. This includes:

- Classification methods (LDA, logistic regression) — these are Week 9+
- Clustering methods (k-means, GMM) — these are Week 9+
- Nonlinear methods (t-SNE, UMAP, kernel PCA) — these are Week 11+
- Any method or visualization type not explicitly taught in class

Creative combinations and extensions of Weeks 1–8 methods are encouraged. But importing techniques from outside the course scope will not earn credit—even if they are good techniques.

If you're unsure whether something is within scope, ask your TA or the instructor *before* submission.

Submission Requirements

Submit the following on Quercus by **March 15, 2026, 11:59 PM**.

1. Written Report (PDF)

Length: 6–10 pages (excluding references), **Figures:** 4–6 well-designed visualizations

We recommend using LaTeX with a standard article format (e.g., \documentclass{article} with 11pt or 12pt font and 1-inch margins). Other formats are acceptable as long as they follow these guidelines.

Your report should be a coherent narrative, not a list of outputs. Guide the reader through your analysis, explaining what you did, what you found, and what it means. Justify your methodological choices explicitly—don't just report results, explain *why* you made the decisions you did.

Formatting:

- 11pt or 12pt font, reasonable margins (1 inch recommended)
- Figures embedded in the text near their discussion
- Informative figure captions that could stand alone
- Page numbers
- References where appropriate (dataset sources, methods citations)

Page limits are enforced. Submissions under 6 pages will receive a score of 0. Content beyond page 10 will be ignored during grading (references excluded).

2. Code (Jupyter Notebook or Script)

Languages: Python (preferred), R, or Julia

Your code must be:

- **Reproducible:** Running it with only path changes should recreate your analysis
- **Organized:** Clear structure with sections matching your report
- **Commented:** Explain non-obvious steps

Submit as .ipynb, .py, .Rmd, .R, or .jl file (or similar file types).

Code must be reproducible. It should run with only minor path changes to the data files. If your code has significant errors or missing files, the TA will consult with you to try to resolve issues, then consult with the instructor before applying any penalty.

3. AI Attribution (if applicable)

If you used generative AI tools (ChatGPT, Claude, GitHub Copilot, etc.), you must provide attribution describing how they were used. This can be a paragraph in your report, a separate text file, or comments in your code.

Examples:

- "Used Claude to debug factor_analyzer package errors and suggest rotation comparisons."
- "Used ChatGPT to help interpret loading patterns and name factors."

Failure to attribute AI assistance when used is an academic integrity violation and will be referred to the instructor.

Late Policy

10% penalty per day, maximum 3 days late. After 3 days, submissions will not be accepted without accommodation.

Evaluation

Your project will be evaluated using the detailed rubric on Quercus. The rubric has three components:

Component	Weight	Sub-dimensions
Visualization & Communication	30%	Figure Design & Readability (15), Writing Quality & Narrative (15)
Methodological Rigor	40%	Method Selection & Justification (20), Assumption Testing & Robustness (20)
Interpretation of Structure	30%	Analysis Execution (15), Interpretation Depth (15)

Each sub-dimension is scored 0–4, where 4 represents exceptional work that is deliberately difficult to achieve. Most competent submissions score in the 2–3 range. See the rubric for specific criteria at each level.

Compared to Project 1, the emphasis shifts from visualization and communication (which were 70% of P1) to methodological rigour (40% of P2). You are expected to apply the visualization and communication skills developed in P1 while demonstrating deeper analytical judgment.

FAQ

Do I need to use every method from Weeks 1–8?

Use the methods appropriate for your data, but justify your choices. EDA and PCA are expected for every submission. At least one of FA or ICA is required. MVN assessment and assumption testing are expected. CCA and multivariate regression are expected where the data structure supports them. If you omit a method, explain why it is not appropriate for your data.

Do I need to use both FA and ICA?

No, although you must use at least one. If you use only FA, discuss whether ICA might be appropriate and why or why not. If you use only ICA, do the same for FA. Using both and comparing them is valuable, but not mandatory.

What about CCA? My data doesn't have obvious variable blocks.

CCA requires a meaningful partition of variables into two blocks. Some P2 datasets have natural block structure (e.g., subscales, sensor types); others don't. If CCA doesn't apply to your data, describe why not. If it does apply, it's a valuable part of your analysis.

My data doesn't seem to have a clean factor structure. Is that a problem?

Not at all. Some datasets have cleaner factor structure than others. If your data's factor structure is ambiguous or weak, discuss *why*—this demonstrates understanding. Explaining why FA doesn't work well (with evidence

from assumption tests and diagnostic statistics) is just as valuable as applying it successfully.

My KMO is high, but the FA results don't make sense. What should I do?

High KMO indicates shared variance, but it can be inflated by multicollinearity. Check your correlation matrix for extremely high correlations. Investigate whether PCA or ICA might be more appropriate for your data. Discussing why FA doesn't work well is valuable.

How should I assess robustness?

Consider: Does the factor structure change if you extract the factors? Does varimax vs promax rotation change interpretation substantially? Do results replicate across random subsamples? Are bootstrap confidence intervals on key loadings wide or narrow? You don't need all of these—pick the robustness checks most relevant to your analysis and report the results.

How many factors should I extract?

Use multiple criteria: parallel analysis, scree plot, and interpretability. These may disagree—discuss the tradeoffs. Don't just pick the number that parallel analysis gives; consider whether the resulting factors are interpretable and meaningful.

Should I use varimax or promax rotation?

Varimax is expected as your primary rotation. If you believe factors should be correlated (oblique), you can additionally apply promax and compare. Justify your choice.

Can I use packages we haven't covered in class?

Yes, as long as you can explain what they do. The factor_analyzer Python package is recommended for FA. sklearn.decomposition.FastICA is recommended for ICA.

What if I discover something unexpected in the data?

Report it. Discovering and explaining anomalies, unexpected factor structures, or data quality issues is valuable. Don't hide problems—investigate and discuss them.

Getting Help

- **Office Hours:** See the syllabus for times and locations
- **Piazza:** Post questions (check if yours has been asked first)
- **Dataset questions:** Direct to your assigned Lead TA for that dataset

Start early. Factor interpretation takes more thought than PCA—allow time for iteration. And remember: justifying *why* you made a choice is as important as the choice itself.

Project 2 Rubric

Criteria	Ratings					Pts
Code Reproducibility Your code must be reproducible — it should run with only minor path changes to the original data files.	**0 pts** **Fail** Code has significant errors, missing files, or cannot be reasonably executed			**0 pts** **Pass** Code runs or would run with trivial fixes		0 pts
AI Attribution If you used generative AI tools (ChatGPT, Claude, Copilot, etc.), you must provide attribution describing how the tools were used.	**0 pts** **Fail** AI clearly contributed but no attribution provided			**0 pts** **No Marks** Attribution provided (if AI was used) or no AI was used		0 pts
Collaboration Statement If work was done in substantial collaboration with others, you must provide a statement describing the nature of the collaboration	**0 pts** **Fail** Collaboration was not acknowledged even though it clearly occurred			**0 pts** **No Marks** Collaboration was outlined (if it occurred) or work was done independently		0 pts
Figure Design & Readability Are the figures well-designed, easy to read, and	**15 pts** **Excellent** Publication-quality figures with informative captions that stand alone. Font	**11 pts** **Good** Well-designed figures with clear labels, legends, and readable fonts. Good	**7 pts** **Fair** Functional figures that convey information. Design is basic	**4 pts** **Poor** Figures present but hard to read. Missing labels, legends, or titles. Poor colour	**0 pts** **Unacceptable** Missing figures, or figures so poorly designed that they fail to communicate.	15 pts

Criteria	Ratings					Pts
accompanied by informative captions?	sizes are appropriate throughout. Colours chosen thoughtfully. No overlapping text or clutter. Axes clearly labelled with units. Captions explain what the figure shows, highlight the key finding, and note why it matters. Would not look out of place in a journal article.	colour choices. Captions explain what the figure shows and note key observations. Minor issues that don't impede understanding (e.g., slightly small text, one caption that describes but doesn't interpret).	with some readability issues (small fonts, dense layouts). Default software styling with minimal customization. Captions describe what the figure shows, but don't highlight key findings.	choices. Captions are perfunctory labels only (e.g., "Figure 2: Factor loadings").	Missing or incorrect captions.	
Writing Quality & Narrative Does the report read as a coherent narrative that introduces the data, explains the analysis, and guides the reader through the findings?	**15 pts** **Excellent** Exceptional writing that guides the reader through the analysis as a coherent story. Data is clearly introduced with domain context — a reader unfamiliar with the dataset understands what it contains and why it matters. Each section builds on the previous. Transitions explain connections. Findings explained in accessible terms. The reader understands not just what was done but why each step matters.	**11 pts** **Good** Clear, logical organization with a good data introduction. Methods and results are explained. Transitions between sections present. The reader can follow the analytical thread and understand the dataset. Some explanations could be clearer or more connected.	**7 pts** **Fair** Readable but feels like a sequence of outputs rather than a narrative. Data description is present but lacks depth. Sections are loosely connected. Results stated but not always explained. The reader must work to understand the analytical logic.	**4 pts** **Poor** Difficult to follow. Minimal data introduction — jumps into analysis. Poor organization or missing sections. Results listed without context. Technical jargon unexplained.	**0 pts** **Unacceptable** Incoherent or severely disorganized. No data description. Major logical gaps. Cannot determine what analysis was performed or what was found.	15 pts
Method Selection &	**20 pts** **Excellent**	**15 pts** **Good**	**10 pts** **Fair**	**5 pts** **Poor**	**0 pts** **Unacceptable**	20 pts

Criteria	Ratings					Pts
Justification Are analytical choices explicitly justified? Does the student explain why they chose each method and why alternatives were or weren't appropriate?	Every major analytical choice is explicitly justified with clear reasoning grounded in the methods' assumptions and the properties of the data. Alternatives are considered, and their inclusion or exclusion is explained. Key decisions (which methods, how many latent dimensions, rotation, preprocessing, etc.) are justified using multiple criteria with discussions of tradeoffs when criteria disagree. Demonstrates understanding that method choice requires careful judgment.	Most analytical choices are justified. Explains why primary methods were selected. Key decisions are justified with at least one clear criterion. The reader understands why these methods were chosen for this data, though some choices could be better motivated.	Methods are applied but choices are not well justified. May state what was done without explaining why. Key decisions are made, but reasoning is thin or relies on a single criterion. Methods seem chosen by default rather than by reasoning about the data.	Methods applied with little or no justification. Key decisions made without explanation. The reader cannot tell why these methods were selected over alternatives. May apply inappropriate methods without recognizing the mismatch.	No justification for any choices. Methods are missing or applied in ways that suggest a fundamental misunderstanding.	
Assumption Testing & Robustness Are assumptions tested rather than assumed? Are results assessed for sensitivity to methodological choices?	**20 pts** **Excellent** Assumptions are systematically tested, and results are interpreted, not just reported. Distributional and structural assumptions are assessed, with clear discussions of violations and their implications for the analysis. Method appropriateness assessed with appropriate caveats. Results	**15 pts** **Good** Key assumptions tested and interpreted. Method appropriateness assessed. At least one robustness or sensitivity check was performed. Results are interpreted, not just reported. Minor gaps in coverage.	**10 pts** **Fair** Some assumption testing present but perfunctory. Tests may be run, but results are not interpreted beyond stating numbers. No robustness checks — a single analysis is presented without a sensitivity assessment. Assumptions are treated as	**5 pts** **Poor** Minimal assumption testing. May mention assumptions in passing, but without formal tests or interpretation. No robustness assessment. Assumptions appear to be taken for granted.	**0 pts** **Unacceptable** No assumption testing. Methods applied with no assessment of whether assumptions hold. No evidence that the student considered whether the methods are appropriate for this data.	20 pts

Criteria	Ratings					Pts
	assessed for robustness through sensitivity analyses (e.g., varying key parameters, comparing alternative approaches, resampling, or quantifying uncertainty). Discusses how robust or fragile the findings are.		checkboxes rather than informing the analysis.			
Analysis Execution Is the analysis technically correct and properly implemented?	**15 pts** **Excellent** Technically impeccable execution. Data appropriately preprocessed (e.g., standardization, missing values). Methods implemented correctly with proper settings. All relevant diagnostics and outputs computed and examined. Numerical results are correct and consistent across the report. No errors or questionable implementation choices.	**11 pts** **Good** Correct execution with appropriate preprocessing. Methods were conducted properly. Key outputs and diagnostics are present and correct. Minor issues at most (e.g., a rounding inconsistency, one missing diagnostic).	**7 pts** **Fair** Analysis is present and mostly correct, but with some gaps in outputs or diagnostics. Preprocessing adequate. Meets minimum requirements, but implementation is basic.	**4 pts** **Poor** Analysis attempted but with notable technical issues. May have errors in preprocessing or implementation. Key outputs are missing. Implementation suggests a limited understanding of the methods.	**0 pts** **Unacceptable** Analysis missing, fundamentally flawed, or results clearly incorrect. Major conceptual errors (e.g., misinterpreting what outputs represent, confusing distinct quantities).	15 pts
Interpretation Depth Does the analysis reveal meaningful latent structure and interpret	**15 pts** **Excellent** Insightful interpretation that reveals genuine understanding. Factors or components given	**11 pts** **Good** Solid interpretation of structure. Factors/Components are named meaningfully based on loading patterns. Makes some connections	**7 pts** **Fair** Basic interpretation present. Reports a number of factors/components and lists high-loading variables, but the interpretation is	**4 pts** **Poor** Minimal interpretation. Reports numerical results (communalities, loadings) without	**0 pts** **Unacceptable** No interpretation, or an interpretation that is fundamentally wrong. Misunderstands	15 pts

Criteria	Ratings					Pts
what it represents?	meaningful domain-specific names justified by loading patterns. Interpretation connects to domain knowledge and earlier EDA findings. Compares what different methods reveal (PCA vs FA, FA vs ICA, rotated vs unrotated, etc.) and discusses what each comparison tells us. Explains what the latent structure means for understanding the data. Discusses limitations and what the analysis cannot reveal. Could teach someone about this dataset by reading this section.	between EDA and FA/ICA results and compares at least two methods. The reader understands the key latent dimensions discovered.	superficial. Findings were stated but not synthesized across methods.	explaining their meaning. No domain connection. No method comparison. The reader learns what was computed but not what was discovered.	what FA/ICA reveals. Conclusions unsupported by the analysis.	
						Total Points: 100